

Quantitative Studies of Science: from sociology and history to simulations

Carlo Debernardi



July 23 2026 - Bochum

Summary

Scientific publishing for dummies

Towards science as an industry

Simulations of scientific research

Part 1

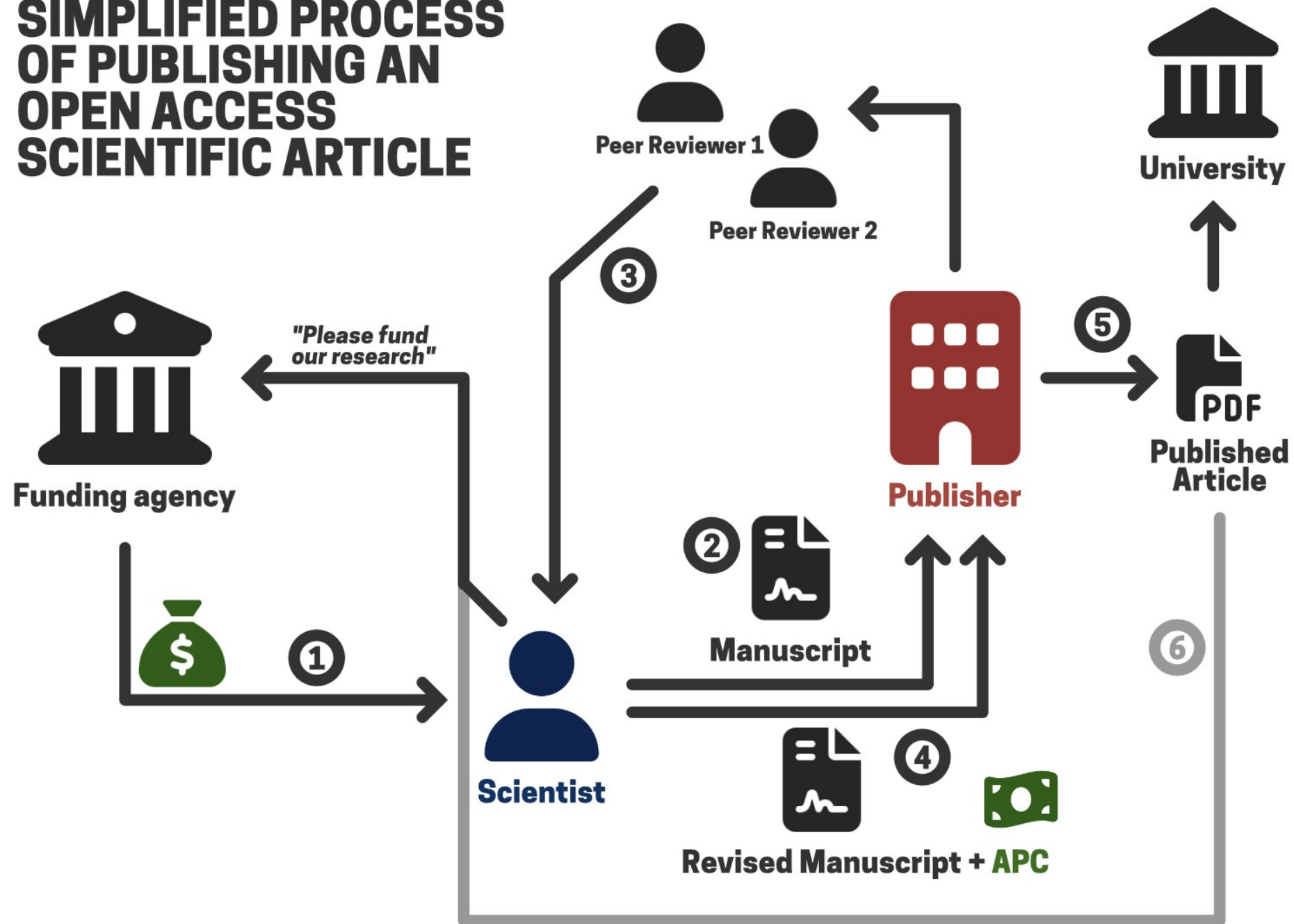
Scientific publishing for dummies

«But, **for science to be advanced**, it is not enough that fruitful ideas be originated or new experiments developed or new problems formulated or new methods instituted. **The innovations must be effectively communicated to others**. That, after all, is what we mean by a contribution to science-something given to the common fund of knowledge. In the end, then, science is a socially shared and socially validated body of knowledge. For the development of science, **only work that is effectively perceived and utilized by other scientists, then and there, matters.**»

(Merton 1979)

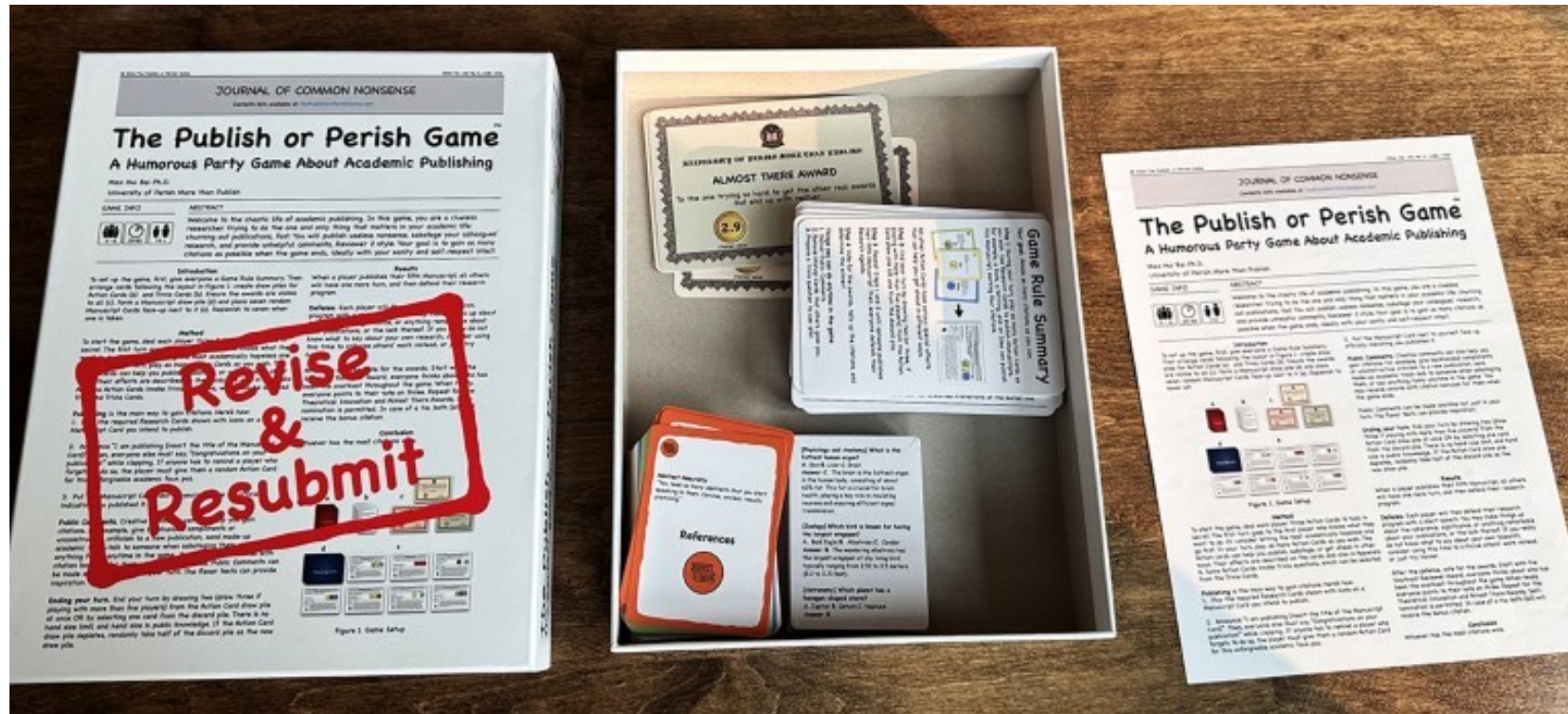
But... Is the scientific literature an
accurate archive of established knowledge?

SIMPLIFIED PROCESS OF PUBLISHING AN OPEN ACCESS SCIENTIFIC ARTICLE



«Publish or perish»

Incentives towards **volume** and **impact** of publications
Salient because **career** and **funding** depend on it!





an open access  journal



RESEARCH ARTICLE

The strain on scientific publishing

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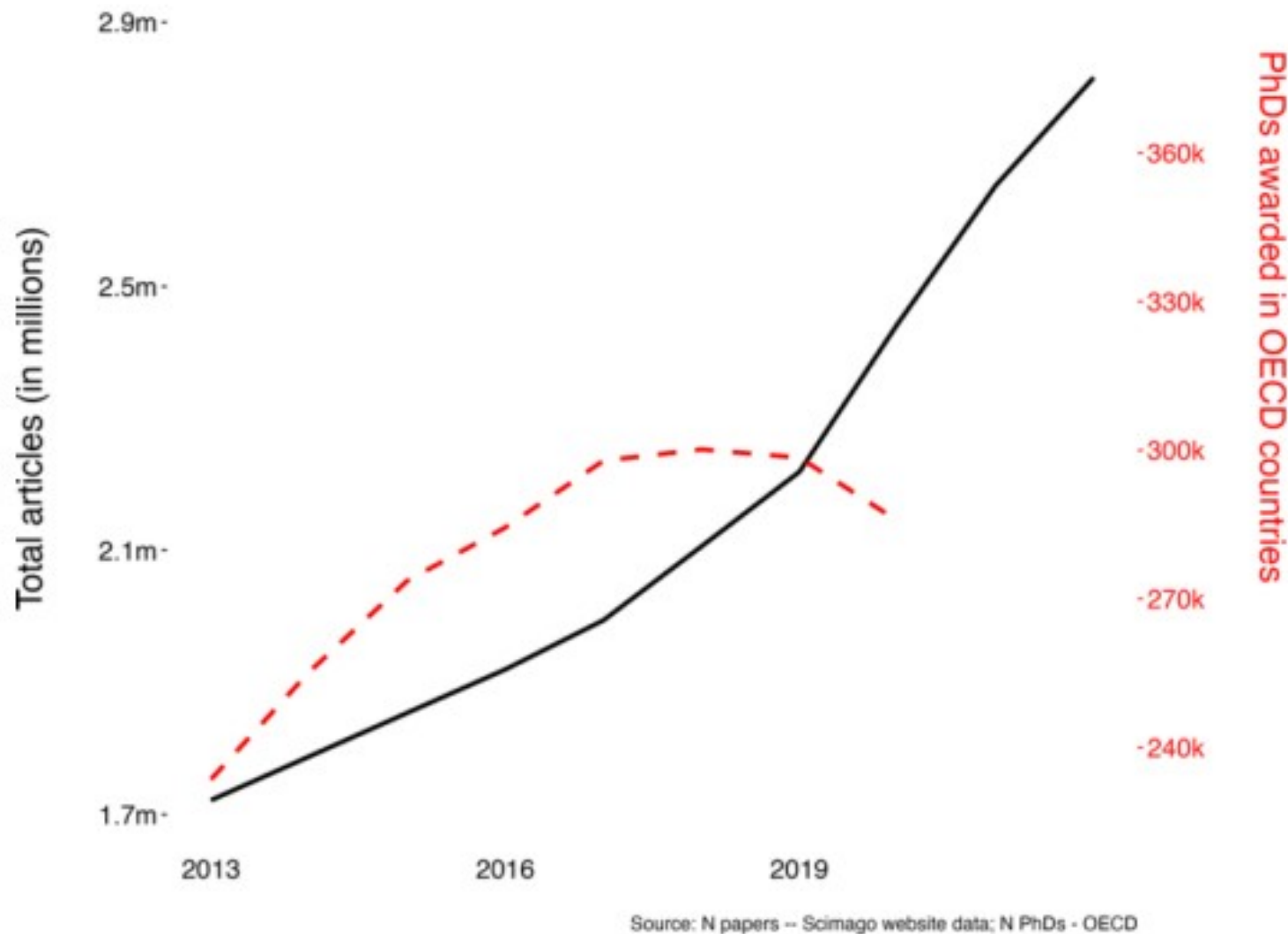
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Keywords: article processing charge, impact factor, open access, Scopus, Web of Science

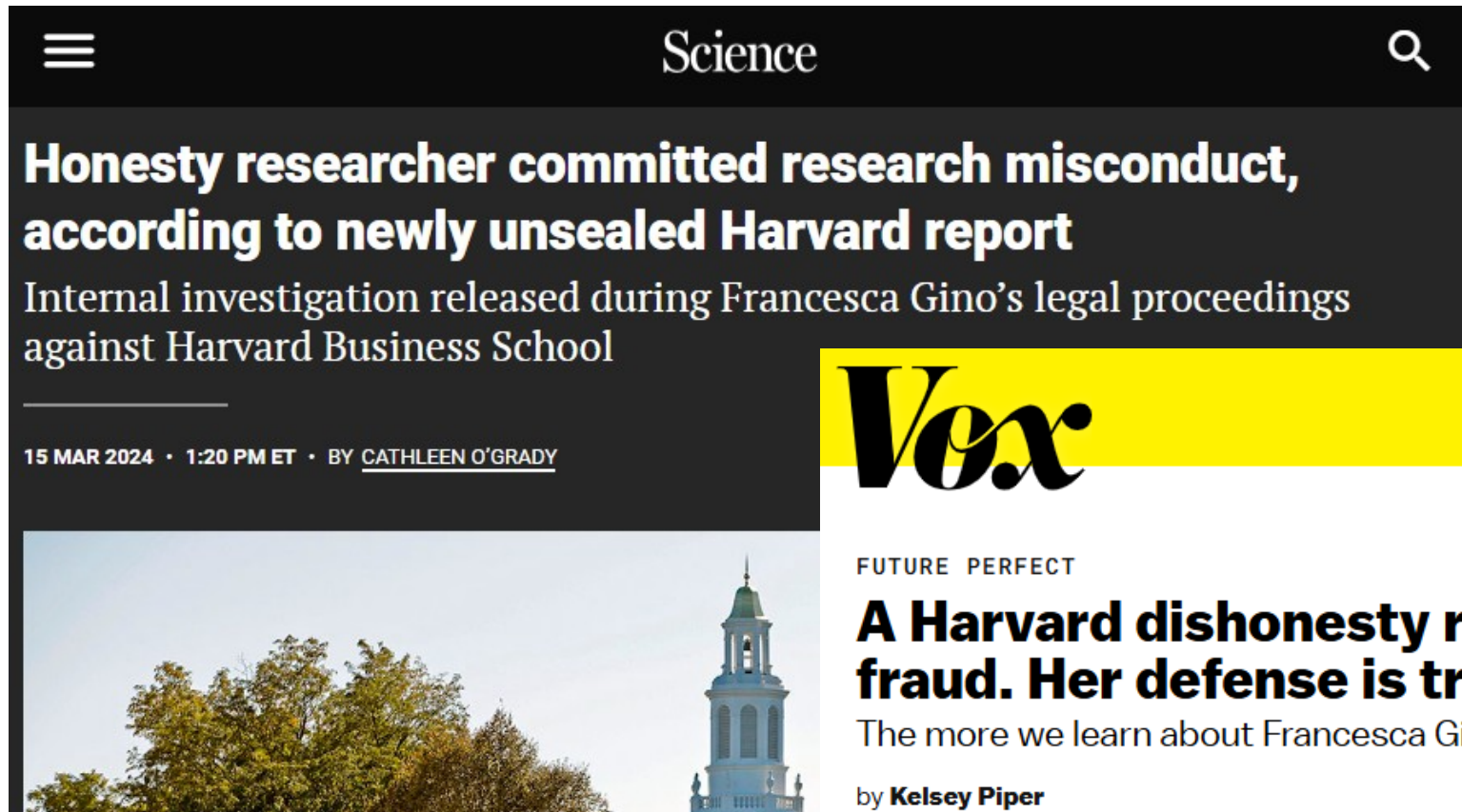
«Academic publishing has a problem. The last few years have seen an **exponential growth in the number of peer-reviewed journal articles**, which has **not been matched by the training of new researchers** who can vet those articles. Editors are reporting **difficulties in recruiting qualified peer reviewers**, and scientists are **overwhelmed by the volume of new articles** being published. We will call this problem “the strain on scientific publishing.”»



«The problem of locating pertinent research literature and the problem of authors' wanting their work to be noticed and used are symmetrical: the vastly increased bulk of publication stiffens the competition between papers for such notice.»

(Merton 1979)

Scientific fraud and misconduct



<https://www.vox.com/future-perfect/24107889/francesca-gino-lawsuit-harvard-dishonesty-researcher-academic-fraud>

<https://www.science.org/content/article/honesty-researcher-committed-research-misconduct-according-newly-unsealed-harvard>



Fraud
Plagiarism
Citation gaming
Predatory publishing
Paper mills
Salami slicing
QRPs
“Not even wrong”
Fashion chasing
Exploration vs exploitation

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OPEN SCIENCE

rsos.royalsocietypublishing.org

Research



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<http://dx.doi.org/10.1098/rsos.160384>

The natural selection of bad science

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Poor research design and data analysis encourage false-positive findings. Such poor methods persist despite perennial calls for

«This paper argues that **some of the most powerful incentives in contemporary science actively encourage, reward and propagate poor research methods** and abuse of statistical procedures. We term this process the natural selection of bad science to indicate that **it requires no conscious strategizing nor cheating on the part of researchers**. Instead, it arises from the positive selection of methods and habits that lead to publication.»

The rise of Metascience

Open access, freely available online

Essay

Why Most Published Research Findings Are False

John P. A. Ioannidis

Article | [Open access](#) | Published: 09 November 2023

RETRACTED ARTICLE: High replicability of newly discovered social-behavioural findings is achievable

[John Protzko](#) , [Jon Krosnick](#), [Leif Nelson](#), [Brian A. Nosek](#), [Jordan Axt](#), [Matt Berent](#), [Nicholas Buttrick](#), [Matthew DeBell](#), [Charles R. Ebersole](#), [Sebastian Lundmark](#), [Bo MacInnis](#), [Michael O'Donnell](#), [Hannah Perfecto](#), [James E. Pustejovsky](#), [Scott S. Roeder](#), [Jan Walleczek](#) & [Jonathan W. Schooler](#)

[Nature Human Behaviour](#) **8**, 311–319 (2024) | [Cite this article](#)

55k Accesses | **19** Citations | **479** Altmetric | [Metrics](#)



This article was [retracted](#) on 24 September 2024

<https://doi.org/10.1371/journal.pmed.0020124>

<https://doi.org/10.1038/s41562-023-01749-9>

The weaponization of the crisis narrative



45 47

← PRESIDENTIAL ACTIONS

Restoring Gold Standard Science

Executive Orders | May 23, 2025

WELCOME TO THE
GOLDEN AGE!

Sign up for news and updates
from the Trump White House.

Your email address

JOIN NOW



Total Number of
Disrupted Grants
5,760



Number of
Reinstated Grants
4,509



Current Loss from
Disrupted Grants
\$533M

<https://www.whitehouse.gov/presidential-actions/2025/05/restoring-gold-standard-science/>

Data only for NIH & AHRQ accessed in July 2026 from: <https://grant-witness.us/>

The case against science is straightforward: much of the scientific literature, perhaps half, may simply be untrue. Afflicted by studies with small sample sizes, tiny effects, invalid exploratory analyses, and flagrant conflicts of interest, together with an obsession for pursuing fashionable trends of dubious importance, science has taken a turn towards darkness.

[...]

Can bad scientific practices be fixed? **Part of the problem is that no-one is incentivised to be right.**

(Horton 2015)



How did it come to this?

Part 2

Towards science as an industry

From Skilled Craftsmanship to Science as an Industry

«Like other institutions, the institution of science has developed an elaborate system for allocating rewards to those who variously live up to its norms. Of course, this was not always so. **The evolution of this system has been the work of centuries, and it will of course never be finished.**»

[...]

«The echo of these complaints [about unrewarded efforts] still reverberates in the halls of universities and scientific societies, but chiefly with regard to material rather than honorific rewards. With the growth and professionalization of science, the system of honorific rewards has become diversely elaborated, and apparently at an accelerated rate.»

(Merton 1979)

The world of science in the early nineteenth century was dominated by **gentlemanly specialists**. The days of the universal virtuoso or polymathic savant in science were over. Scientific endeavor was divided into a set of relatively discrete fields, each with practitioners, institutions, and journals to match.

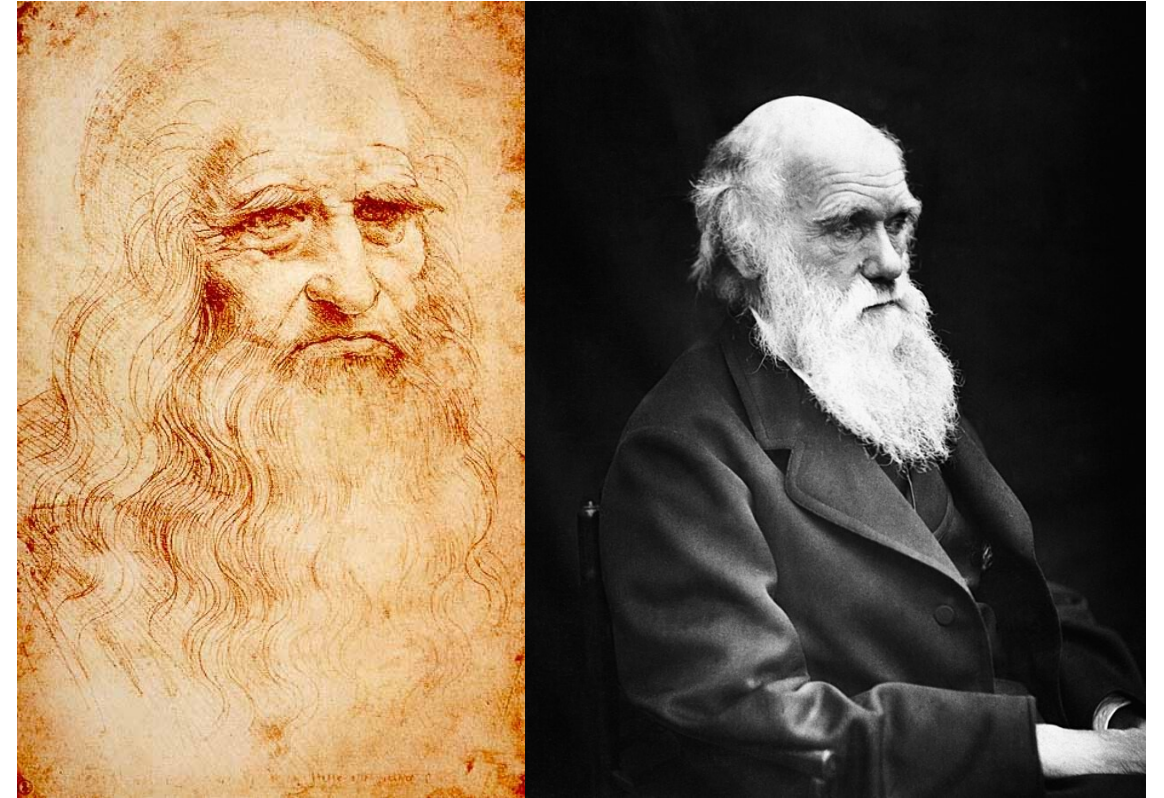
[...]

Although the degree of specialization was only moderate by modern standards, most of the scientific achievements of the early nineteenth century were unmistakably the work of **specialists**.

[...]

Many of the leaders of science at this period, particularly but not only in England, were also unmistakably **gentlemen**.

(Rudwick, 1985)



Scientific Societies: Arenas of Gentlemanly Debate

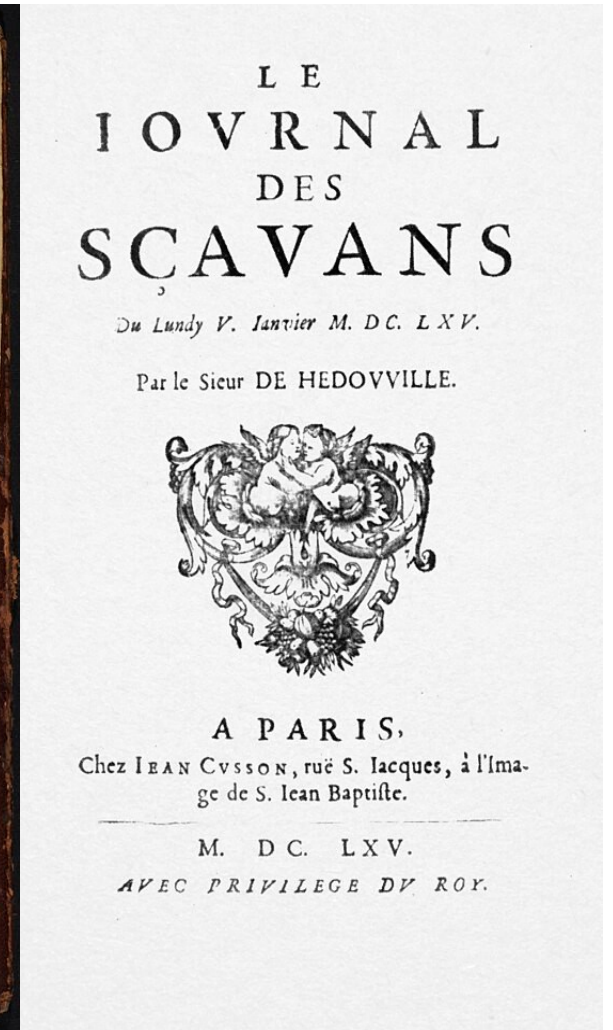
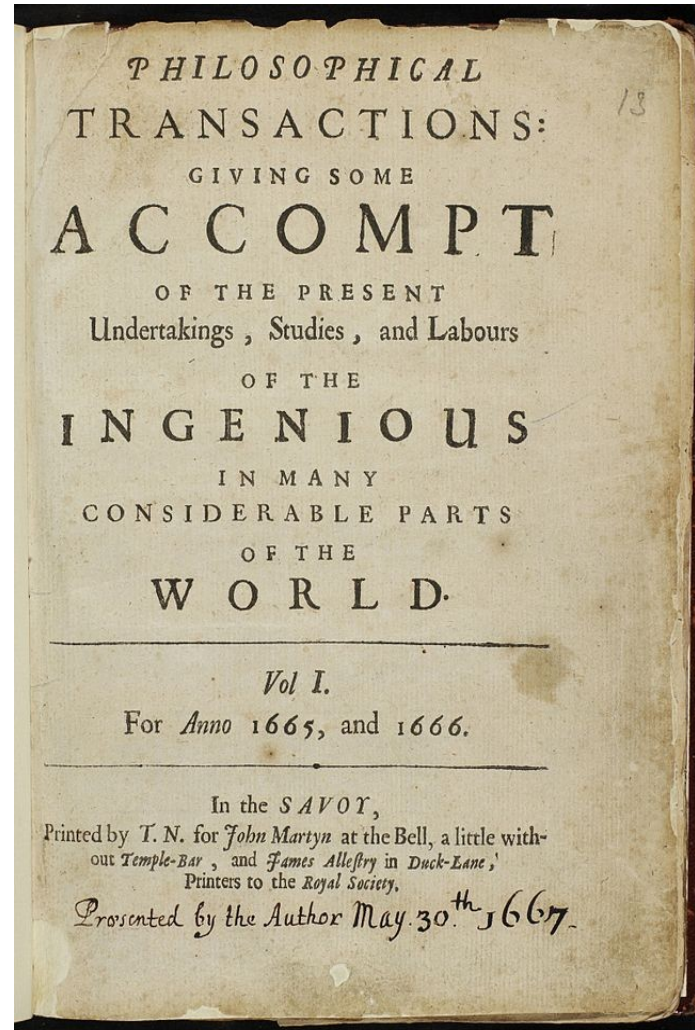


Fig. 2.1. A contemporary sketch of the Geological Society of London in session. Note the parliamentary arrangement of the seating. The leading members—De la Beche is easily recognizable by his spectacles—face each other on the front benches, while less important figures are crowded behind them, and only faintly sketched in. The president, flanked by the two secretaries, occupies the Speaker's position. Specimens for discussion are on the central table (in this sketch, an ichthyosaur's head is just to the right of the president's); visual aids are displayed on the far wall (in this sketch, a large traverse section of a mountain range).

The birth of peer review?

«[*Philosophical Transactions* was] as much the ancestor of the abstract or review journal, or of the scientific news and book reviews now found in the front half of *Nature*, as it is of the peer-reviewed research journal»
(Fyfe et al. 2022)

Peer review used sparingly and as an **editorial support tool**



By the end of the nineteenth century, the transition from science as a calling to science as a job had at most just begun. [...] So at the beginning of the twentieth century the identity of the scientist was radically unstable. **To be a scientist was still something of a calling but it was becoming something of a job.**

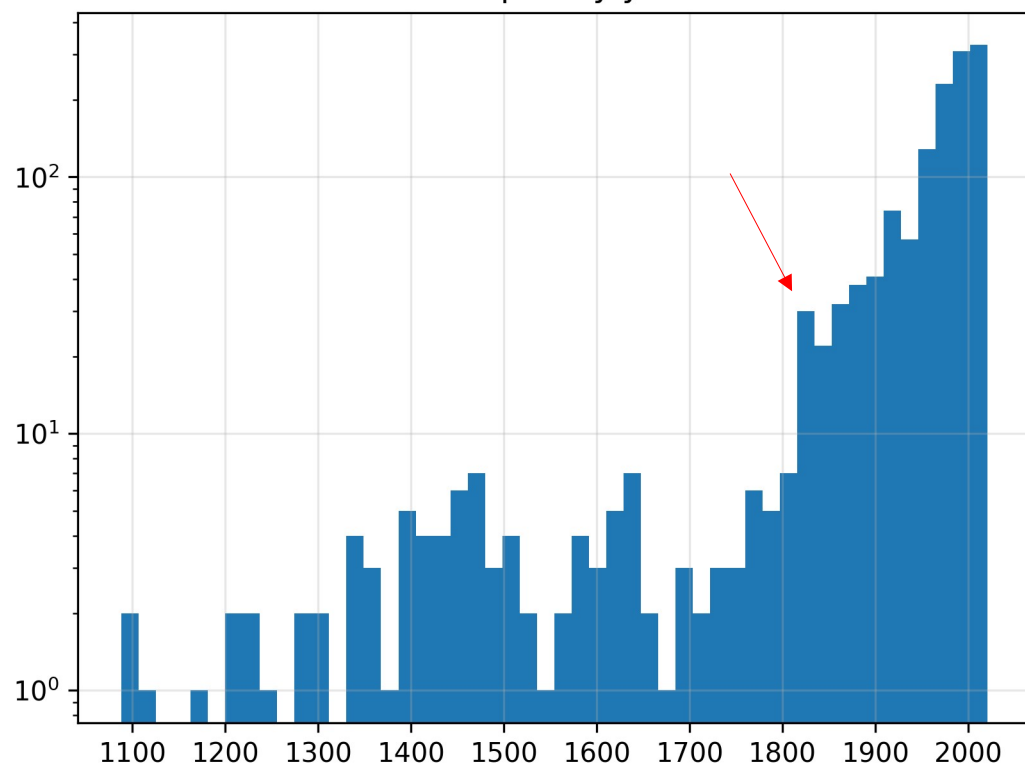
(Shapin 2008)

On 29 November **1924**, the British scientific journal Nature—then in its fiftyfifth year of publication—printed a letter to the editor from the respected physicist Norman R. Campbell. Campbell had a simple request: that Nature discard the terms “man of science” and “scientific worker” and, in their place, begin using the word **scientist**.

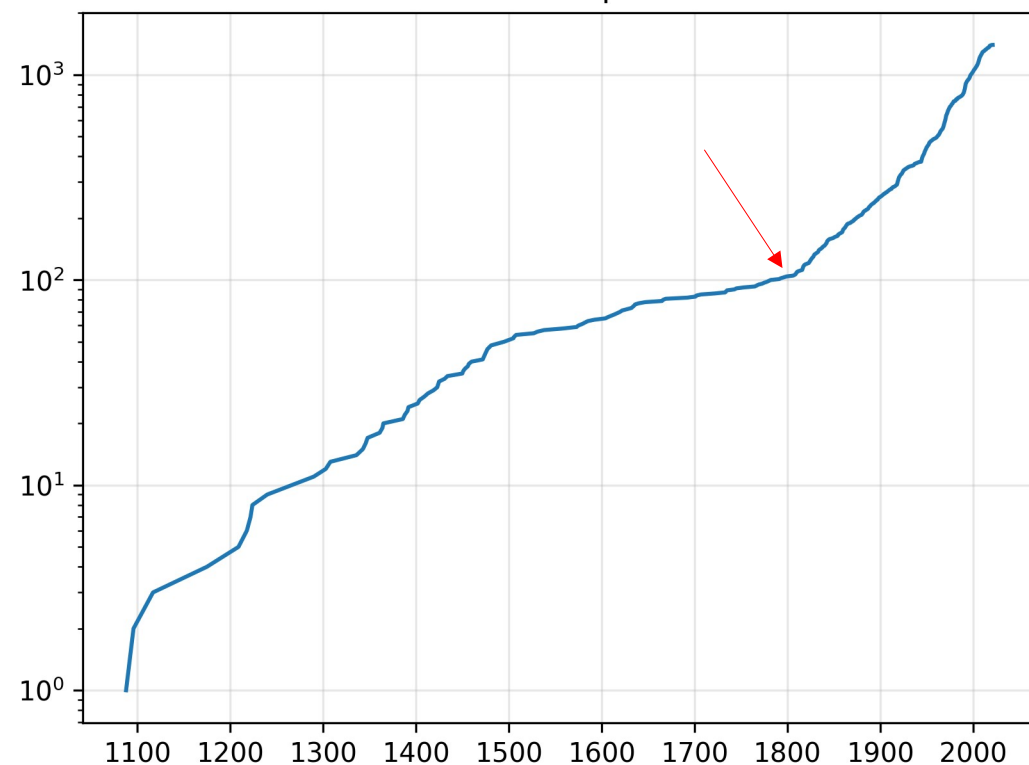
(Baldwin 2015)

Individual professionalization and institutional formalization

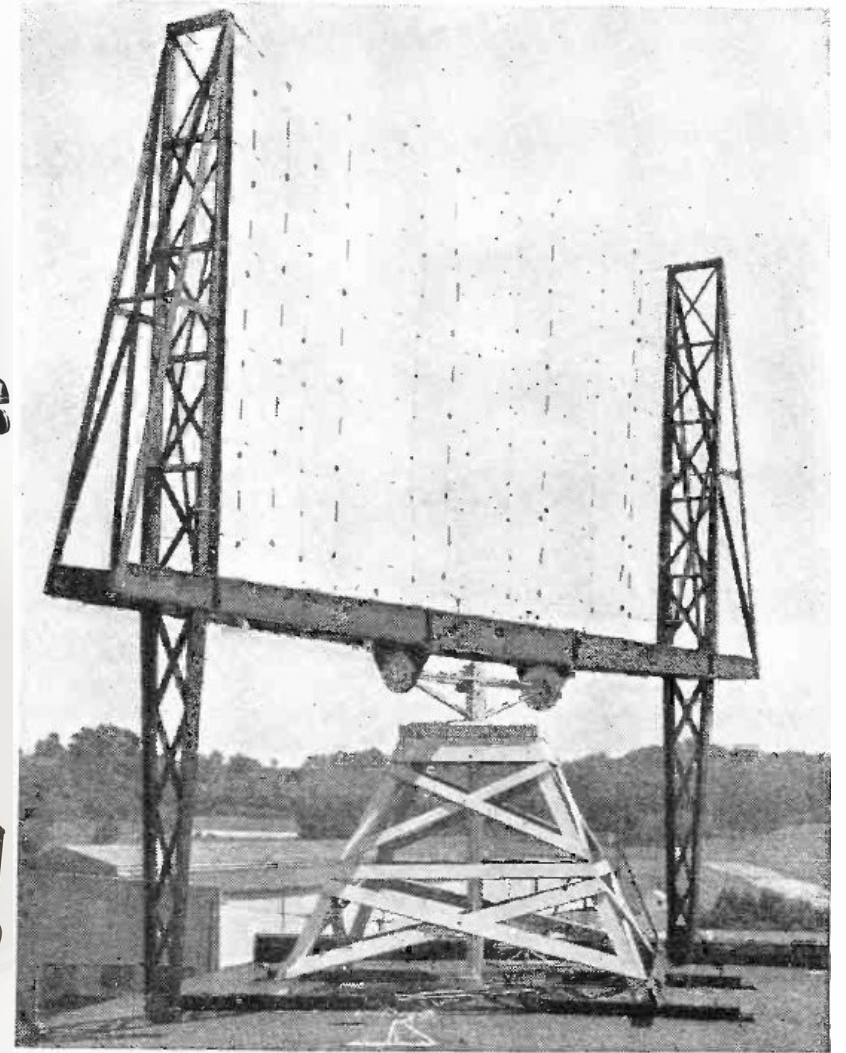
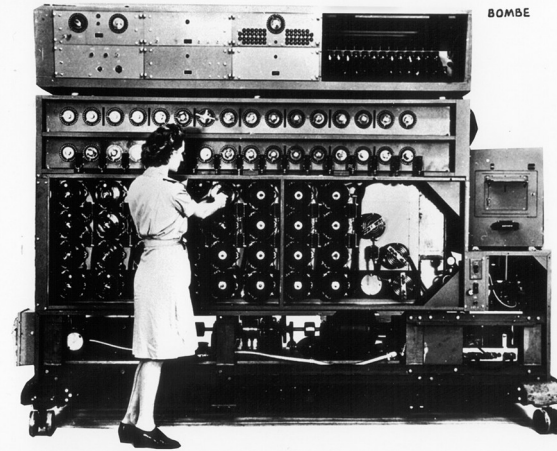
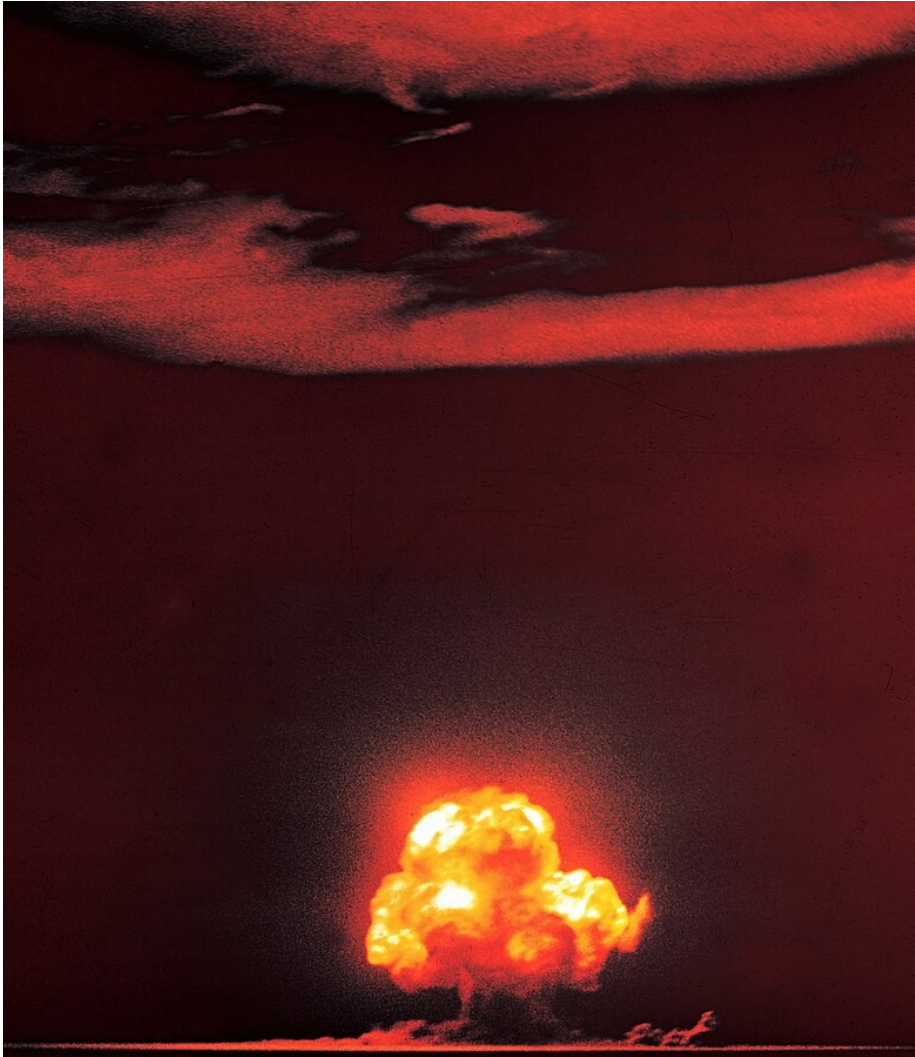
Universities in europe - by year of foundation



Universities in europe - over time



War time applications: from little to big science



War time applications: from little to big science

The **mobilization of American science during the Second World War** – especially in the Manhattan Project and in the construction of radar, but spreading across much of the scientific landscape – propelled a generation of academic scientists into a world that was largely unfamiliar to them: the experience of **large-scale organization**; of **teamwork**; of **interdisciplinary project-oriented research**; of **unlimited resources** and severely **limited time**; of close contact with the sorts of people – especially **the military and the commercial worlds** – they had not known much about and, after the end of the war and the beginning of the Cold War, the experience – for some of them – of **political power**.

(Shapin 2008)

Science— The Endless Frontier

A Report to the President on a Program
for Postwar Scientific Research

by Vannevar Bush
Director of the Office of Scientific
Research and Development
July 1945

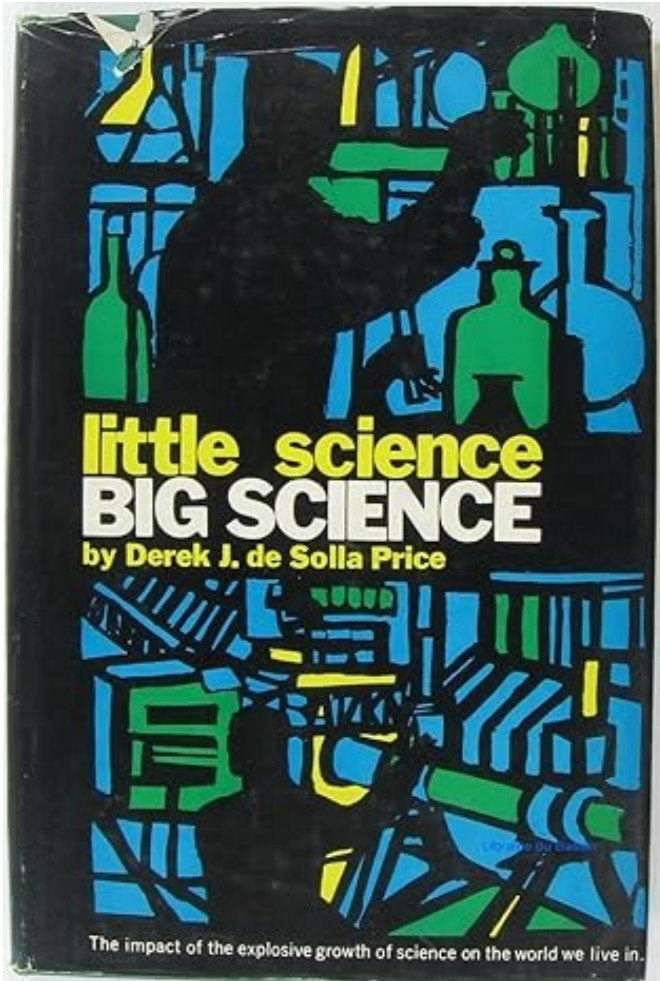
--- SUMMARY OF THE REPORT ---

- SCIENTIFIC PROGRESS IS ESSENTIAL
- For the War Against Disease
- For Our National Security
- And for the Public Welfare
- WE MUST RENEW OUR SCIENTIFIC TALENT
- Including Those in Uniform
- **THE LID MUST BE LIFTED**
- A PROGRAM FOR ACTION

--- INTRODUCTION ---

- Scientific Progress Is Essential
- **Science Is a Proper Concern of Government**
- Government Relations to Science - Past and Future
- **Freedom of Inquiry Must Be Preserved**

Bibliometrics: descriptive or normative?



Citation Indexes for Science

A New Dimension in Documentation
through Association of Ideas

Eugene Garfield

Metaphorically speaking, **citations are frozen footprints** on the landscape of scholarly achievement; footprints **which bear witness to the passage of ideas**. From footprints it is possible to deduce direction; from the configuration and depth of the imprints it should be possible to construct a picture of those who have passed by, whilst the distribution and variety furnish clues as to whether the advance was orderly and purposive.

(Cronin 1981)

Scientific Autonomy, Public Accountability, and the Rise of “Peer Review” in the Cold War United States

Melinda Baldwin, *American Institute of Physics*

«Peer review’s perceived failures may have their roots in the gap between **modern expectations** of refereeing and the more **modest functions** it was **initially designed to fulfill.**»

Throughout the **nineteenth century and into much of the twentieth**, external referee reports were considered an **optional part of journal editing or grant making**. The idea that refereeing is a requirement for scientific legitimacy seems to have arisen first in the **Cold War United States**. In the 1970s, in the wake of a series of attacks on scientific funding, American scientists faced a dilemma: there was increasing pressure for science to be accountable to those who funded it, but scientists wanted to ensure their continuing influence over funding decisions. Scientists and their supporters cast expert refereeing—or “peer review,” as it was increasingly called—as the **crucial process that ensured the credibility of science as a whole**. Taking funding decisions out of expert hands, they argued, would be a corruption of science itself. This public elevation of peer review both reinforced and spread the belief that only peer-reviewed science was scientifically legitimate.



The rise of *New Public Management*

- Limit government spending
- Steering at a distance: market-like dynamics

Bind career and funding to
quality and efficiency, but how?

RAE/REF (UK): peer review and bibliometrics
as **centralised evaluation** tools

But consider Goodhart's law: "When a measure becomes a target, it ceases to be a good measure"

«**The mores of science** possess a methodologic rationale but they **are binding**, not only because they are procedurally efficient, but **because they are believed right and good**. They are **moral as well as technical** prescriptions. Four sets of institutional imperatives-universalism, communism, disinterestedness, organized skepticism-are taken to comprise the ethos of modern science.»

(Merton 1979)

Four Mertonian norms [\[edit \]](#)

The four Mertonian norms (often abbreviated as the CUDO-norms) can be summarised as:

- communism: all scientists should have common ownership of scientific goods (intellectual property), to promote collective collaboration; secrecy is the opposite of this norm.^[3]
- universalism: scientific validity is independent of the sociopolitical status/personal attributes of its participants.^[4]
- disinterestedness: scientific institutions act for the benefit of a common scientific enterprise, rather than for specific outcomes^[5] or the resulting personal gain of individuals within them.
- organized skepticism: scientific claims should be exposed to [critical scrutiny](#) before being accepted: both in methodology and institutional codes of conduct.^[6]

Things are changing...?



- Preprints
- Open data & code
- Replication exercises
- Preregistration
- Post publication and open peer review
- San Francisco Declaration
- ... and many more!



Is that enough?

- Specialisation → division of labour
- Risk of even stronger bureaucratisation
- What about communities?



Part 3

Simulations of scientific research

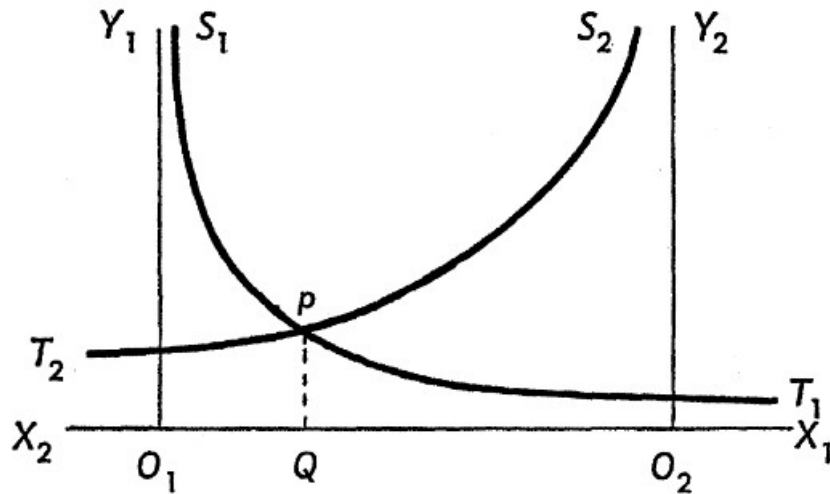
Simulations, but why...?

- Express assumptions without ambiguity
- Make complex dynamics explicit
- Explore how research systems evolve over time
- Identify feedback loops and unintended consequences
- Test “what-if” scenarios safely
- Compare alternative policies, incentives, or interventions
- Reveal emergent patterns and system-level effects
- Support evidence-based decision-making

Back to the origins



new discoveries of the greatest moment. In our day, a thousand chemists, with the most elaborate appliances, are not able to reach results which are comparable in interest with those early ones. All the sciences exhibit the same phenomenon, and so does the course of life. At first we learn very easily, and the interest of experience is very great; but it becomes harder and harder, and less and less worth while, until we are glad to sleep in death.



It is to be remarked that the theory here given rests on the supposition that the object of the investigation is the ascertainment of truth. When an investigation is made for the purpose of attaining personal distinction, the economics of the problem are entirely different. But that seems to be well enough understood by those engaged in that sort of investigation.

The individual point of view

THE JOURNAL OF PHILOSOPHY

VOLUME LXXXVII, NO. 1, JANUARY 1990

Philip Kitcher

THE DIVISION OF COGNITIVE LABOR*

The Role of the Priority Rule in Science

Michael Strevens

Journal of Philosophy 100:55–79, 2003

- Choosing a “theory” as a lottery ticket
- Problem of coordination and individual maximization
- Role of social factors (e.g. priority) in directing the choice

How do we best design social institutions for the advancement of learning? The philosophers have ignored the social structure of science. The point, however, is to change it. - P.K.

Exploration and exploitation

Epistemic Landscapes and the Division of Cognitive Labor*

Michael Weisberg and Ryan Muldoon^{†‡}

- From analytical to agent-based models
- Mavericks vs Followers, optimal mix?
- Many extensions (e.g. Avin 2019)

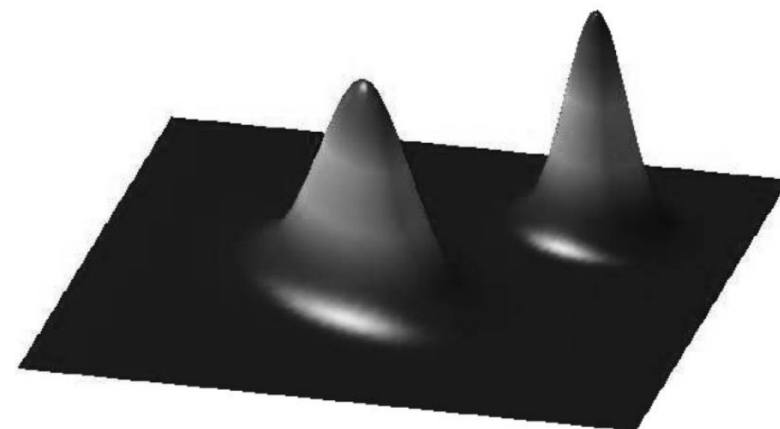
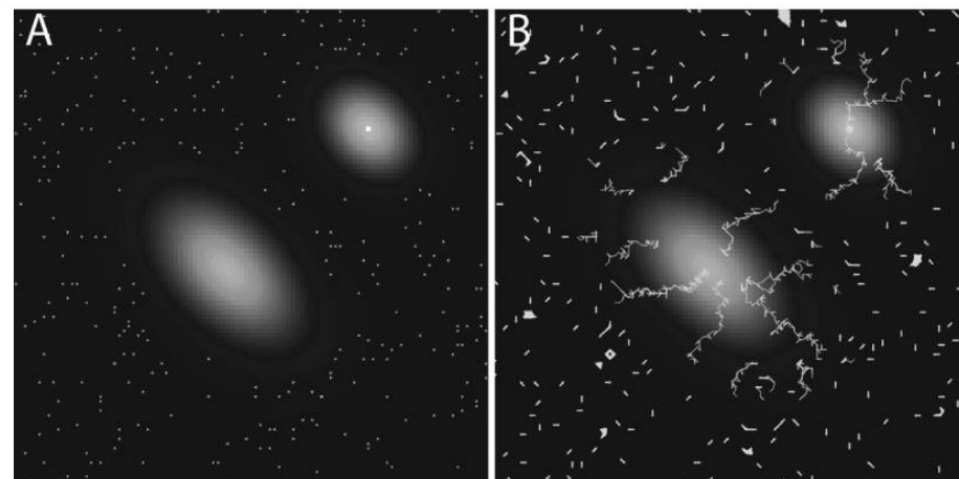


Figure 1. An example epistemic landscape of the form used for the models in this article.



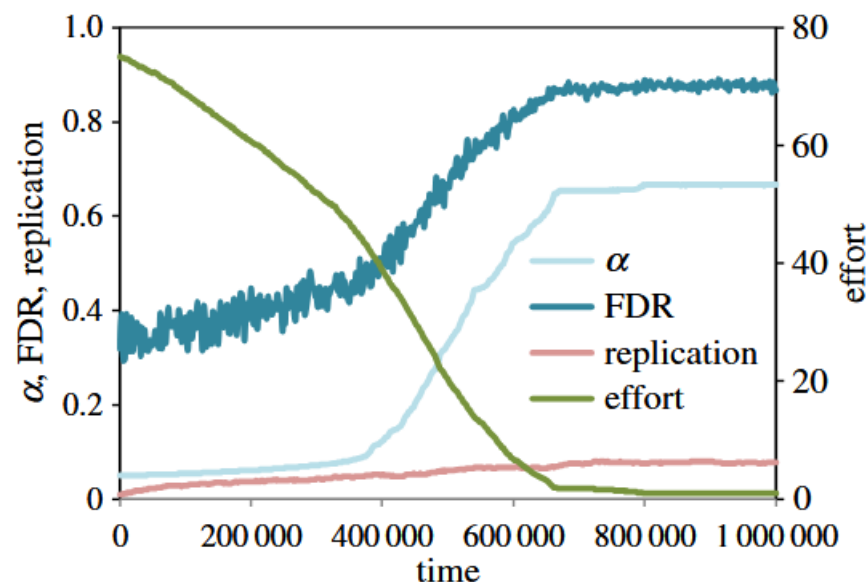
The natural selection of bad science

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- Labs as agents with individual features
- Statistical power for the experiments
- Effort reduces risk but slows down research
- Studies can be replications of previous ones
- In evolutionary version, labs survive based on how many studies they publish
- Even with harsh punishment for no replication False Discovery Rate grows
- In static model replication solves the problem (!)

Without replication, low effort labs have an unambiguous advantage. As the rate of replication increases, the mean pay-off for high effort labs can surpass the mean pay-off for low effort labs, as the former are less likely to be punished. However, the laws of probability dictate that some labs will escape either producing false positives or being caught doing so, and among these the very highest performers will be those who exert low effort. When the top labs have disproportionate influence on funding and graduate student success, **this type of small advantage can cascade to continuously select for lower effort and increasing numbers of false discoveries**, as seen in our evolutionary model.

«[...] some of the most powerful incentives in contemporary science actively encourage, reward and propagate poor research methods and abuse of statistical procedures. [...] The **natural selection of bad science** [...] requires no conscious strategizing nor cheating [...] Instead, it **arises from the positive selection of methods and habits that lead to publication.**»

Smaldino & McElreath (2016)

«How do we best design social institutions for the advancement of learning? The philosophers have ignored the social structure of science. The point, however, is to change it.»

Kitcher (1990)

Thanks!

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